

INFO 5602
Information Visualization

“Choosing an Effective Visual”

We are NOT naturally good at **storytelling** and **presentation structuring**

Presentation = Noise + Signal

- *Clutter/distractions on slides*
- *Stakeholder - irrelevant information included*
- *Stakeholder -irrelevant “design” options*
- *Distracting colors, objects included in presentations*
- ***Adds cognitive load to stakeholders***

Find + Remove

- *Stakeholder-relevant information*
- *Stakeholder-relevant charts*
- *Stakeholder-relevant design options*
- *User-centric design for charts*
- ***Improves effective stakeholder communication***

Find/Create + Boost

Agenda: the key charts you will use professionally

- Simple text
- Tables
- Heatmap
- Graphs
- Points
 - Scatterplot
 - Line graphs
 - Slope graphs
- Bars
- Area charts
- “Pie charts are evil”
- “Never use 3D”

Simple text

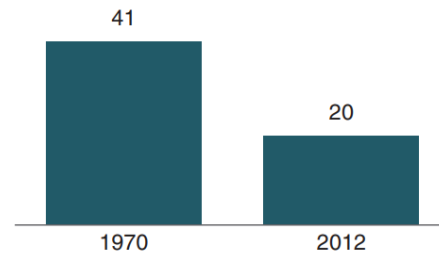
- When you have just a number or two to share, simple text is a strong way to visualize that to stakeholders
- Use the number and make it as prominent as possible – with a few supporting words to make your point
- When you have just a number or two that you want to communicate: **use the numbers directly.**

Simple text presentation example:

“Before” chart

Children with a "Traditional" Stay-at-Home Mother

*% of children with a married
stay-at-home mother with a
working husband*



Note: Based on children younger than 18.
Their mothers are categorized based on
employment status in 1970 and 2012.

Source: Pew Research Center analysis of
March Current Population Surveys
Integrated Public Use Microdata Series
(IPUMS-CPS), 1971 and 2013

Adapted from PEW RESEARCH CENTER

FIGURE 2.2 Stay-at-home moms original graph

“After” chart

20%

of children had a
traditional stay-at-home mom
in 2012, compared to 41% in 1970

FIGURE 2.3 Stay-at-home moms simple text makeover

Tables

- Tables are great for visualizing/communicating to a mixed audience, whose members will each look for their particular row or column of interest
- If you have to use a table to present data, let the design fade into the background – **let the data take center stage**
- Do not let borders or shading compete for (your stakeholder's) attention
 - Use light borders or simply white space to set table elements apart
 - Push **table borders into the background by making them grey or removing them**
- Anything (**other than relevant data**) on your table that competes for your stakeholder's attention **adds noise, which increases cognitive load (and damages effective communication with your stakeholders)**
- Keep tables simple

Tables presentation example:

“Before” chart

Heavy borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

“After” chart options

Light borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Minimal borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

FIGURE 2.4 Table borders

Heatmaps

- Heatmaps = table + visual cues
- A heatmap is a way to visualize data in tabular format, where in place of (or in addition to) the numbers, you use colored cells that **convey the relative magnitude of the numbers to your stakeholders**
- Heatmaps use **color saturation** to:
 - **reduce mental processing (i.e. cognitive load) for your stakeholders,**
 - provide visual cues, helping eyes and brains of your stakeholders more quickly target the potential points of interest

Heatmaps presentation example:

“Before” chart

“After” chart

Table

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Heatmap

LOW-HIGH

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

FIGURE 2.5 Two views of the same data

Graphs

- A well-designed graph will get information across more quickly than a well-designed table Knafllic (2015)
- There are several types of graphs; we will consider the following:
 - Point graphs
 - Line graphs
 - Bar graphs
 - Area graphs

Graphs: Point graphs

- Scatterplots (point graphs) can be useful for showing the relationship between two numeric variables
- Scatterplots tend to be more frequently used in scientific fields – **and can be viewed as complicated by stakeholders not familiar with them**
- Scatterplots can be modified to draw attention to areas of the graph **you want your stakeholders to focus on** (see next slide)

Scatterplot presentation example:

“Before” chart

Cost per mile by miles driven

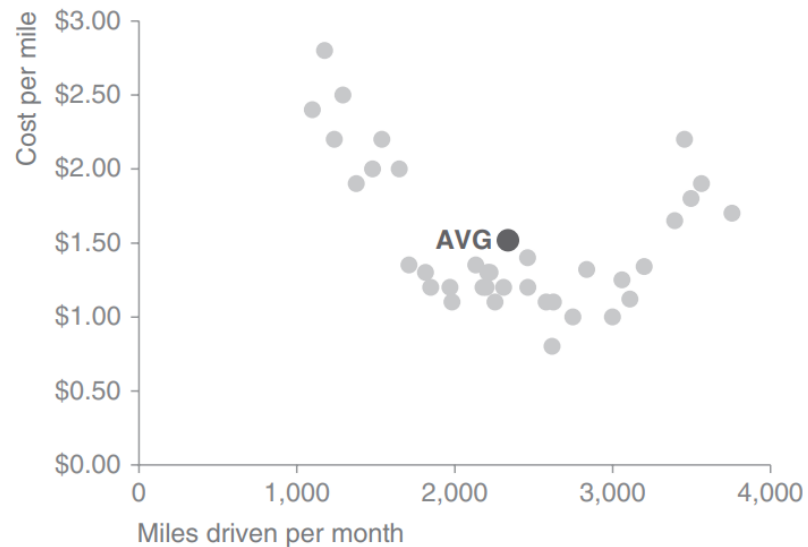


FIGURE 2.6 Scatterplot

“After” chart

Cost per mile by miles driven

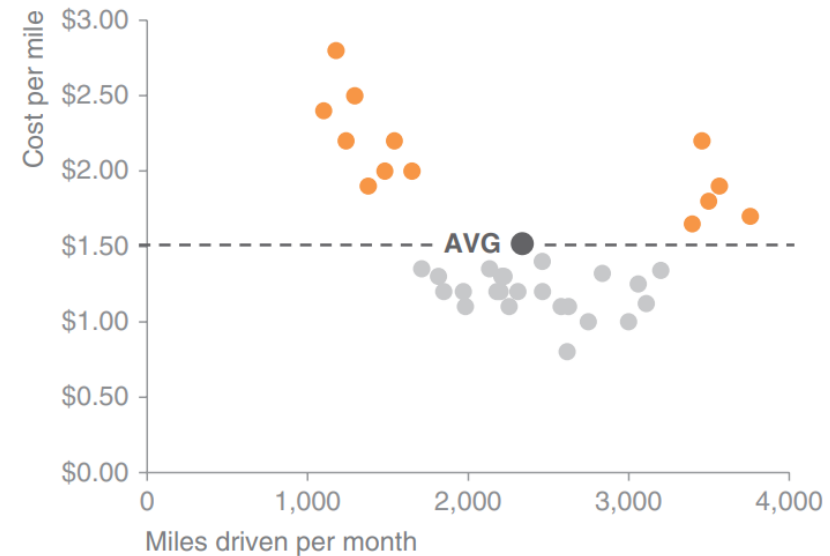


FIGURE 2.7 Modified scatterplot

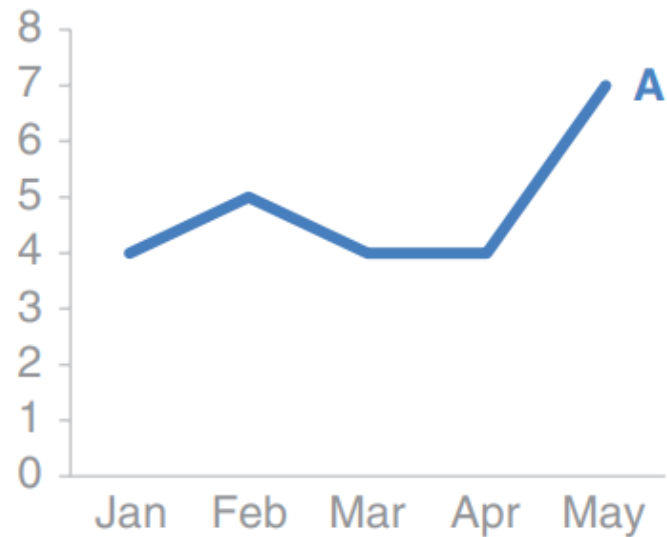
In this example, color is used to highlight to your audience that cost per mile is above average when

- miles driven per month is between 1K and 2K and
- miles driven per month is between 3K and 4K

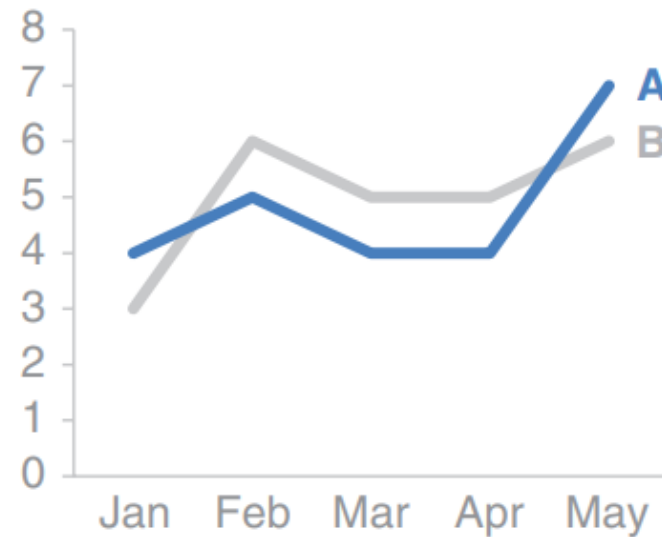
Graphs: Line graphs

- Line graphs are most commonly used **to plot continuous data**
- Continuous data are usually in some unit of time: days, months, quarters, or years
- When you graph time on the horizontal axis of a line graph, the data **must be in consistent intervals.**

Single series



Two series



Multiple series

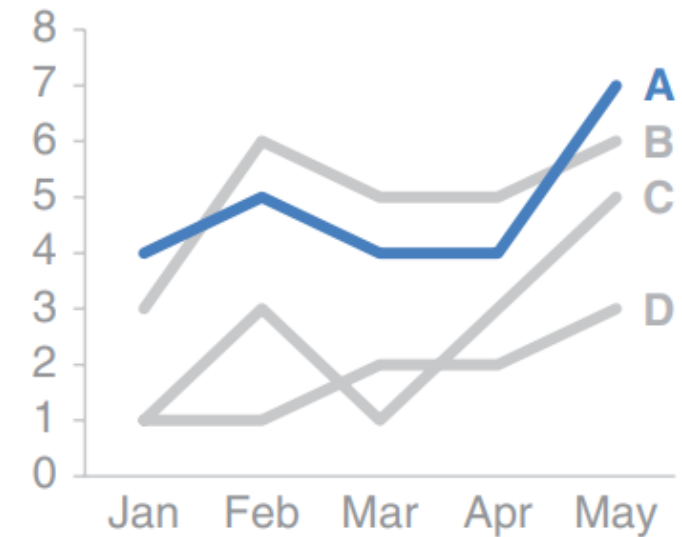


FIGURE 2.8 Line graphs

Graphs: Slope graphs

- Slope graphs can be useful when you have two time periods or points of comparison and **want to quickly show relative increases/decreases or differences across various categories between the two data points**
- Slope graphs can be used to show your stakeholders:
- Absolute numbers
- **Visual increase or decrease** in the rate of change (via the slope or direction) without having to explain what “rate of change” is

Slope graph presentation example:

“Before” chart

Employee feedback over time

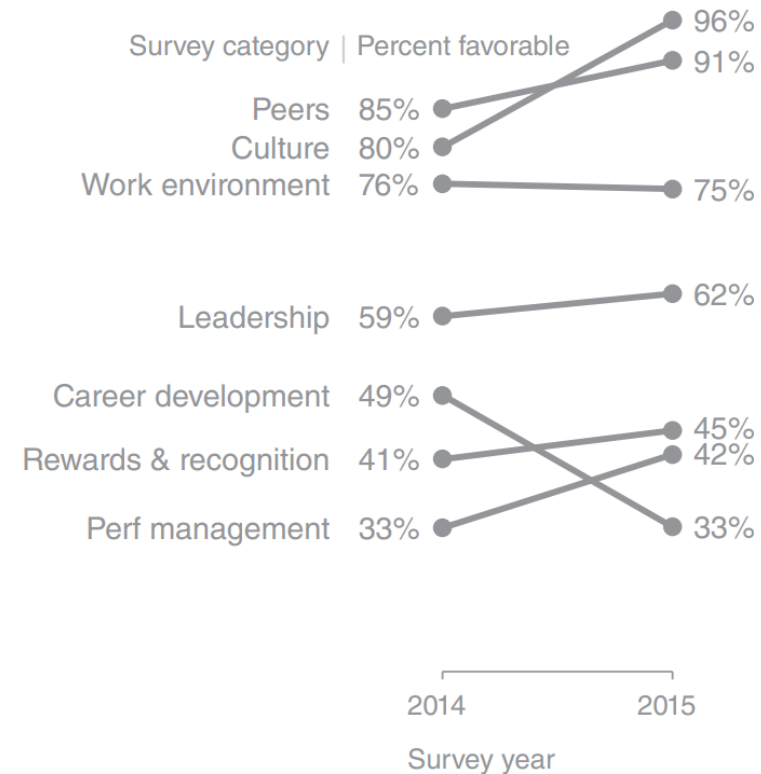


FIGURE 2.10 Slopegraph

Source: Knafllic (2015)

“After” chart

Employee feedback over time

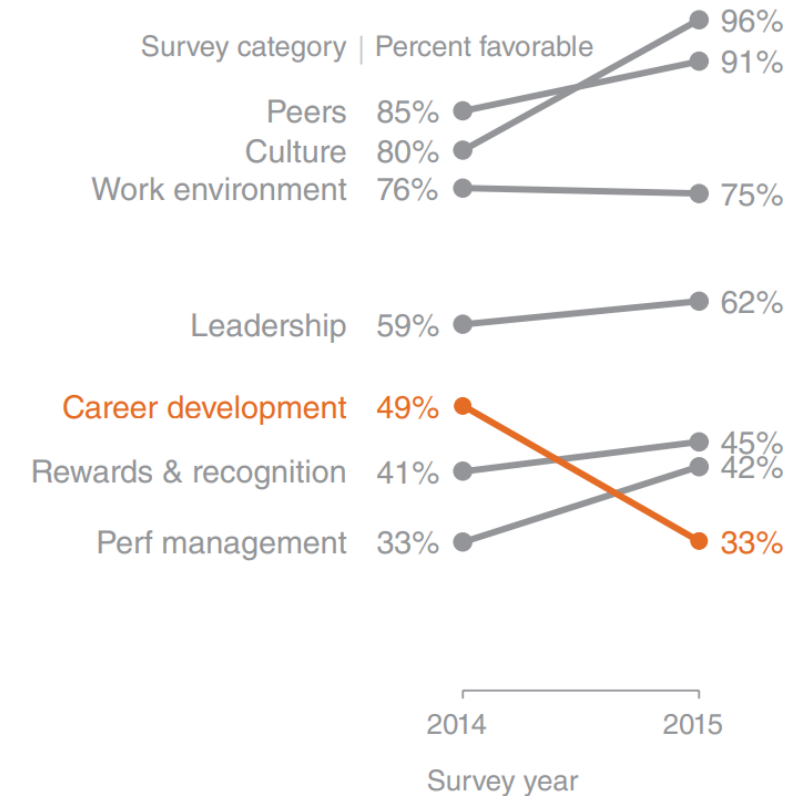


FIGURE 2.11 Modified slopegraph

End of presentation